Similar Triangles

Class 2

How to use this companion. Read, discuss, and refer to the diagrams and questions during class. For every question, draw a rough figure, mark the given measurements, and write every step with its reason in your notebook.

1 What we know from Class 1

Two triangles are **similar** when they have the same shape, though not necessarily the same size. For similar triangles,

- the corresponding angles are equal, and
- the corresponding sides are in the same ratio. This common ratio is the **scale factor**.

The statement $\triangle ABC \sim \triangle PQR$ also tells us which vertices correspond: $A \leftrightarrow P$, $B \leftrightarrow Q$, $C \leftrightarrow R$. So

$$\angle A = \angle P, \quad \angle B = \angle Q, \quad \angle C = \angle R \qquad \text{and} \qquad \frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP}.$$

Exercise 2A

1. $\triangle ABC \sim \triangle PQR$. Write the angle of $\triangle PQR$ that is equal to $\angle B$, and the side of $\triangle PQR$ that corresponds to AC. Then complete $\dfrac{AB}{PQ} = \dfrac{}{QR} = \dfrac{}{\quad}$.

2. A triangular kolam pattern has sides 6 cm, 8 cm and 10 cm. An enlarged copy of the same shape has its shortest side 9 cm. Find the other two sides of the copy.

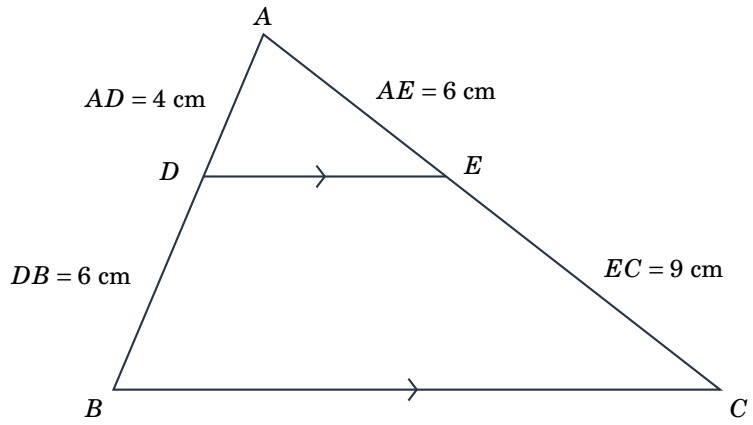
3. $\triangle DEF \sim \triangle XYZ$, with $DE = 12$ cm, $XY = 8$ cm and $YZ = 10$ cm. Find EF.

4. Arjun says, “The triangles with sides 3, 4, 5 cm and 5, 6, 7 cm are similar, because every side of the second triangle is 2 cm longer.” Is he right? Explain.

5. $\triangle ABC \sim \triangle PQR$, with $AB = 6$ cm, $BC = 8$ cm and $PQ = 9$ cm. A student writes $QR = \dfrac{8 \times 6}{9} = \dfrac{16}{3}$ cm. Find the mistake and find the correct QR.

2 A parallel line divides two sides in the same ratio

In $\triangle ABC$, the segment DE is drawn parallel to BC, with D on AB and E on AC.



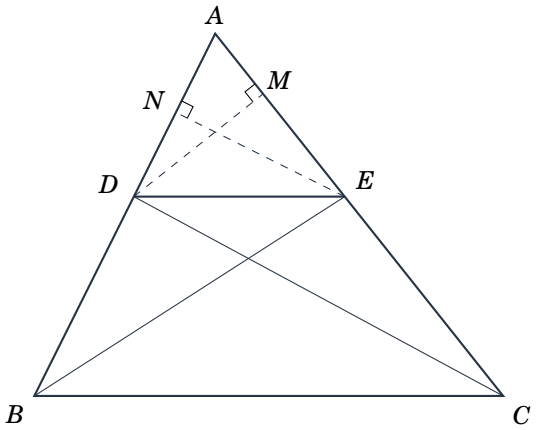
Activity. (a) Find $\dfrac{AD}{DB}$ and $\dfrac{AE}{EC}$. What do you notice?

(b) Find $\dfrac{AD}{AB}$ and $\dfrac{AE}{AC}$. What do you notice?

(c) Draw any triangle ABC in your notebook. Use a set square to draw a line parallel to BC that cuts AB at D and AC at E. Measure AD, DB, AE and EC, and compare $\dfrac{AD}{DB}$ with $\dfrac{AE}{EC}$.

The parallel line cuts the two sides in the same ratio. Measuring suggests this, but a drawing cannot show that it happens in *every* triangle. We need a proof.

Theorem 1 (Basic Proportionality Theorem)**.** *If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.*



Given: In $\triangle ABC$, $DE \parallel BC$, with D on AB and E on AC.
To prove: $\dfrac{AD}{DB} = \dfrac{AE}{EC}$.
Construction: Join BE and CD. Draw $EN \perp AB$ and $DM \perp AC$.

Proof. Write ar for the area of a triangle.

Triangles ADE and BDE have bases AD and DB on the same line, and the same height EN. So

$$\frac{\text{ar}(ADE)}{\text{ar}(BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN} = \frac{AD}{DB}. \tag{1}$$

Triangles ADE and DEC have bases AE and EC on the same line, and the same height DM. So

$$\frac{\text{ar}(ADE)}{\text{ar}(DEC)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC}. \tag{2}$$

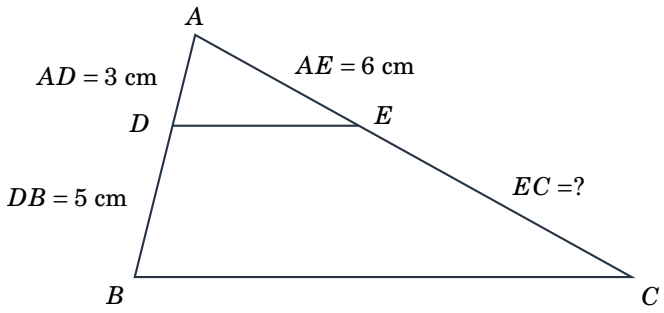
Triangles BDE and DEC are on the same base DE and between the same parallels DE and BC. So

$$\text{ar}(BDE) = \text{ar}(DEC). \tag{3}$$

By (3), the left sides of (1) and (2) are equal. Therefore $\frac{AD}{DB} = \frac{AE}{EC}$. □

The theorem is also called **BPT**. From now on we use only this name.

Example 2.1. In $\triangle ABC$, $DE \parallel BC$, with D on AB and E on AC. $AD = 3$ cm, $DB = 5$ cm and $AE = 6$ cm. Find EC.



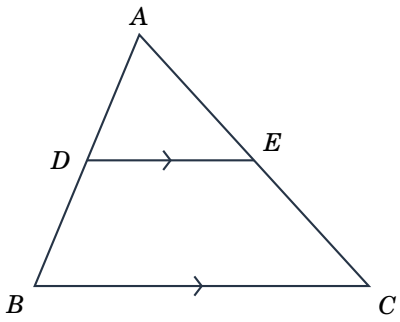
Given: $DE \parallel BC$, $AD = 3$ cm, $DB = 5$ cm, $AE = 6$ cm.
To find: EC.

Solution.

$$\frac{AD}{DB} = \frac{AE}{EC} \qquad \text{(BPT, since } DE \parallel BC\text{)}$$
$$\frac{3}{5} = \frac{6}{EC}$$
$$EC = \frac{6 \times 5}{3} = 10.$$

Answer: $EC = 10$ cm.

Example 2.2. In $\triangle ABC$, a line parallel to BC meets AB at D and AC at E. Prove that $\frac{AD}{AB} = \frac{AE}{AC}$.



Given: In $\triangle ABC$, $DE \parallel BC$.
To prove: $\frac{AD}{AB} = \frac{AE}{AC}$.

Proof.

$$\frac{AD}{DB} = \frac{AE}{EC} \qquad \text{(BPT)}$$

$$\frac{DB}{AD} = \frac{EC}{AE} \qquad \text{(take the reciprocal of each side)}$$

$$\frac{DB}{AD} + 1 = \frac{EC}{AE} + 1 \qquad \text{(add 1 to each side)}$$

$$\frac{DB + AD}{AD} = \frac{EC + AE}{AE}$$

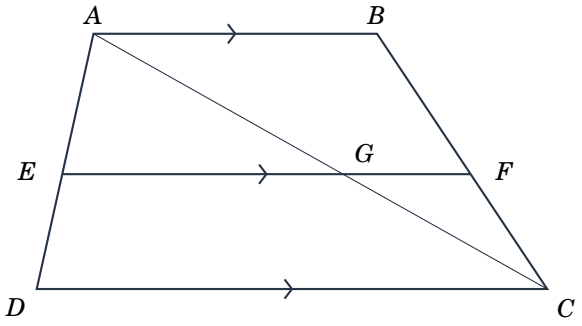
$$\frac{AB}{AD} = \frac{AC}{AE} \qquad (AD + DB = AB \text{ and } AE + EC = AC)$$

$$\frac{AD}{AB} = \frac{AE}{AC}. \qquad \text{(take the reciprocal again)}$$

□

So BPT can be used in either form: part to part, $\frac{AD}{DB} = \frac{AE}{EC}$, or part to whole side, $\frac{AD}{AB} = \frac{AE}{AC}$.

Example 2.3. $ABCD$ is a trapezium with $AB \parallel DC$. E is a point on AD and F is a point on BC such that $EF \parallel AB$. Prove that $\frac{AE}{ED} = \frac{BF}{FC}$.



Given: $AB \parallel DC$ and $EF \parallel AB$.
To prove: $\frac{AE}{ED} = \frac{BF}{FC}$.
Construction: Join AC. Let it meet EF at G.

Proof. Since $EF \parallel AB$ and $AB \parallel DC$, we have $EF \parallel DC$.

In $\triangle ADC$, $EG \parallel DC$. So

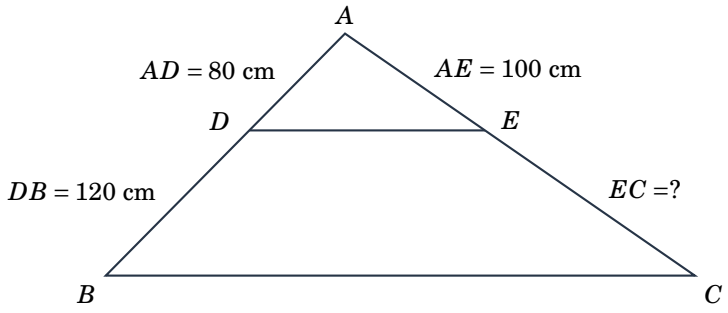
$$\frac{AE}{ED} = \frac{AG}{GC}. \qquad \text{(BPT)}$$

In $\triangle CAB$, $GF \parallel AB$. So

$$\frac{CG}{GA} = \frac{CF}{FB}, \quad \text{that is,} \quad \frac{AG}{GC} = \frac{BF}{FC}. \qquad \text{(BPT, then reciprocals)}$$

Both results equal $\frac{AG}{GC}$. Therefore $\frac{AE}{ED} = \frac{BF}{FC}$. □

Example 2.4. The front of a village house roof is a triangle ABC, with A at the top. A horizontal wooden beam DE is fixed parallel to the base BC, with D on AB and E on AC. $AD = 80$ cm, $DB = 120$ cm and $AE = 100$ cm. Find EC and the length of the slope AC.



Given: $DE \parallel BC$, $AD = 80$ cm, $DB = 120$ cm, $AE = 100$ cm.
To find: EC and AC.

Solution.

$$\frac{AD}{DB} = \frac{AE}{EC} \qquad \text{(BPT, since } DE \parallel BC\text{)}$$

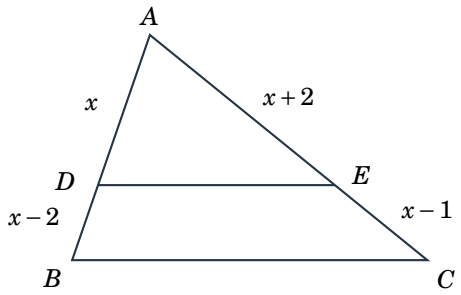
$$\frac{80}{120} = \frac{100}{EC}$$

$$EC = \frac{100 \times 120}{80} = 150.$$

$$AC = AE + EC = 100 + 150 = 250.$$

Answer: $EC = 150$ cm and $AC = 250$ cm.

Example 2.5. In $\triangle ABC$, $DE \parallel BC$, with D on AB and E on AC. $AD = x$ cm, $DB = (x-2)$ cm, $AE = (x+2)$ cm and $EC = (x-1)$ cm. Find x.



Given: $DE \parallel BC$, $AD = x$, $DB = x-2$, $AE = x+2$, $EC = x-1$ (in cm).
To find: x.

Solution.

$$\frac{x}{x-2} = \frac{x+2}{x-1} \qquad \text{(BPT)}$$

$$x(x-1) = (x+2)(x-2) \qquad \text{(cross-multiply)}$$

$$x^2 - x = x^2 - 4$$

$$x = 4.$$

Check: $AD = 4$, $DB = 2$, $AE = 6$, $EC = 3$, and $\frac{4}{2} = \frac{6}{3}$.

Answer: $x = 4$.

Exercise 2B

1. In $\triangle ABC$, $DE \parallel BC$, with D on AB and E on AC. $AD = 6$ cm, $DB = 4$ cm and $AE = 9$ cm. Find EC.

2. In $\triangle PQR$, S is on PQ and T is on PR, with $ST \parallel QR$. $PS = 4$ cm, $PQ = 10$ cm and $PR = 15$ cm. Draw a rough figure and mark the given lengths. Find PT and TR.

3. In $\triangle ABC$, $DE \parallel BC$, with D on AB and E on AC. $AD = x$ cm, $DB = (x+3)$ cm, $AE = (x+1)$ cm and $EC = (x+5)$ cm. Find x. Then find AB and AC.

4. In $\triangle ABC$, $DE \parallel BC$, $AD = 5$ cm, $DB = 10$ cm and $AE = 4$ cm. Ravi writes

$$\frac{AD}{DB} = \frac{EC}{AE}, \qquad \frac{5}{10} = \frac{EC}{4}, \qquad EC = 2 \text{ cm}.$$

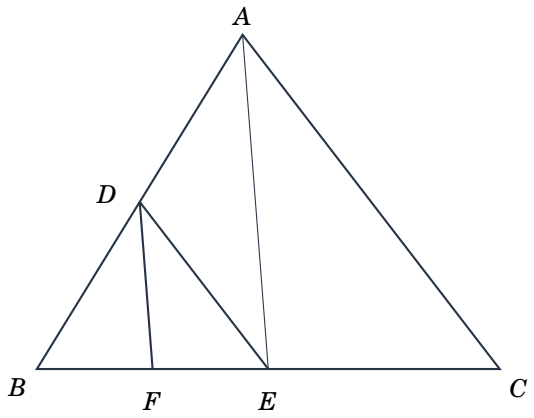
   Find Ravi's mistake. Find the correct EC.

5. In $\triangle ABC$, $DE \parallel BC$. Kiran says, "Because $\dfrac{AD}{DB} = \dfrac{AE}{EC}$, it is also true that $\dfrac{DB}{AB} = \dfrac{EC}{AC}$." Is he right? Explain.

6. Draw a triangle ABC with $AB = 6$ cm and $AC = 9$ cm (choose any angle at A). Mark D on AB with $AD = 2$ cm. Use a set square to draw $DE \parallel BC$, with E on AC. Before you measure, calculate AE. Then measure AE and compare.

7. A triangular plot of land ABC lies beside a road. A fence DE is built parallel to the side BC, with D on AB and E on AC. $AD = 30$ m, $DB = 20$ m and $AC = 75$ m. How far from A, along AC, does the fence meet AC?

8. In $\triangle ABC$, D is on AB and E is on BC, with $DE \parallel AC$. F is on BE with $DF \parallel AE$. Prove that $\dfrac{BF}{FE} = \dfrac{BE}{EC}$.

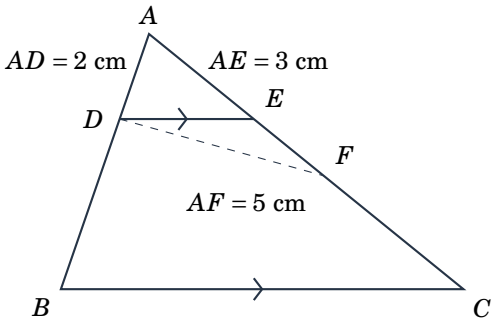


3 Can equal division force a parallel line?

BPT says: if $DE \parallel BC$, then $\dfrac{AD}{DB} = \dfrac{AE}{EC}$. Now ask the reverse question. If D and E divide AB and AC in the same ratio, must DE be parallel to BC?

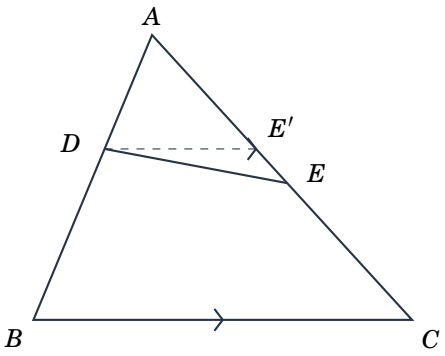
Activity. (a) Draw a triangle ABC with $AB = 6$ cm and $AC = 9$ cm. Mark D on AB with $AD = 2$ cm and E on AC with $AE = 3$ cm. Find $\dfrac{AD}{DB}$ and $\dfrac{AE}{EC}$. Join DE and check with a set square whether $DE \parallel BC$.

(b) On the same triangle, mark F on AC with $AF = 5$ cm. Find $\dfrac{AF}{FC}$. Join DF. Is $DF \parallel BC$?



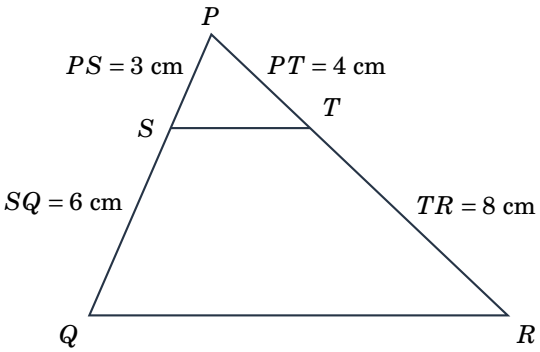
Theorem 2 (Converse of BPT)**.** *If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side.*

Why it works. In $\triangle ABC$, let $\dfrac{AD}{DB} = \dfrac{AE}{EC}$. Suppose DE is *not* parallel to BC. Draw $DE' \parallel BC$, with E' on AC.



By BPT, $\dfrac{AD}{DB} = \dfrac{AE'}{E'C}$. So $\dfrac{AE}{EC} = \dfrac{AE'}{E'C}$. Adding 1 to each side gives $\dfrac{AC}{EC} = \dfrac{AC}{E'C}$. So $EC = E'C$, and E and E' are the same point. Therefore DE is the line DE', and $DE \parallel BC$.

Example 3.1. In $\triangle PQR$, S is on PQ and T is on PR. $PS = 3$ cm, $SQ = 6$ cm, $PT = 4$ cm and $TR = 8$ cm. Is $ST \parallel QR$? Give a reason.



Given: $PS = 3$ cm, $SQ = 6$ cm, $PT = 4$ cm, $TR = 8$ cm.
To find: whether $ST \parallel QR$.

Solution.

$$\frac{PS}{SQ} = \frac{3}{6} = \frac{1}{2}, \qquad \frac{PT}{TR} = \frac{4}{8} = \frac{1}{2}.$$

The two ratios are equal. So $ST \parallel QR$ (converse of BPT).

Answer: Yes, $ST \parallel QR$.

Example 3.2. In $\triangle PQR$, S is on PQ and T is on PR. $PS = 3$ cm, $SQ = 6$ cm, $PT = 5$ cm and $TR = 8$ cm. Is $ST \parallel QR$?

Given: $PS = 3$ cm, $SQ = 6$ cm, $PT = 5$ cm, $TR = 8$ cm.
To find: whether $ST \parallel QR$.

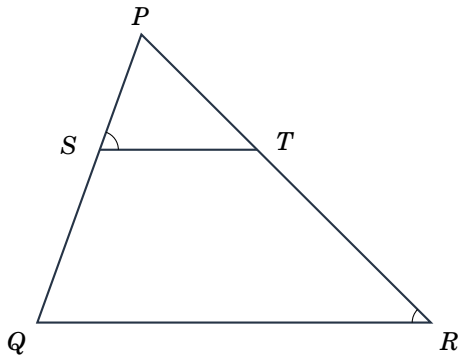
Solution.

$$\frac{PS}{SQ} = \frac{1}{2}, \qquad \frac{PT}{TR} = \frac{5}{8}.$$

The ratios are not equal. If ST were parallel to QR, BPT would make them equal. So ST is not parallel to QR.

Answer: No.

Example 3.3. In $\triangle PQR$, S is on PQ and T is on PR, with $\dfrac{PS}{SQ} = \dfrac{PT}{TR}$ and $\angle PST = \angle PRQ$. Prove that $\triangle PQR$ is isosceles.



Given: $\dfrac{PS}{SQ} = \dfrac{PT}{TR}$ and $\angle PST = \angle PRQ$.
To prove: $\triangle PQR$ is isosceles.

Proof.

$ST \parallel QR$	(converse of BPT)
$\angle PST = \angle PQR$	(corresponding angles, $ST \parallel QR$)
$\angle PST = \angle PRQ$	(given)
$\angle PQR = \angle PRQ$	(both equal $\angle PST$)
$PR = PQ$	(sides opposite equal angles)

So $\triangle PQR$ is isosceles. □

Exercise 2C

1. In $\triangle ABC$, D is on AB and E is on AC. Decide whether $DE \parallel BC$ in each case. Give a reason.

   (a) $AD = 4$ cm, $DB = 6$ cm, $AE = 6$ cm, $EC = 9$ cm.

   (b) $AD = 5$ cm, $DB = 7$ cm, $AE = 6$ cm, $EC = 9$ cm.

2. In $\triangle ABC$, D is on AB and E is on AC. $AB = 12$ cm, $AD = 4$ cm, $AC = 18$ cm and $AE = 6$ cm. Draw a rough figure and mark the given lengths. Is $DE \parallel BC$? Give a reason.

3. In $\triangle ABC$, D is on AB and E is on AC. $AD = 6$ cm, $DB = 9$ cm and $AE = 8$ cm. Find the length EC must have for DE to be parallel to BC.

4. D and E are the midpoints of the sides AB and AC of $\triangle ABC$. Use the converse of BPT to explain why $DE \parallel BC$.

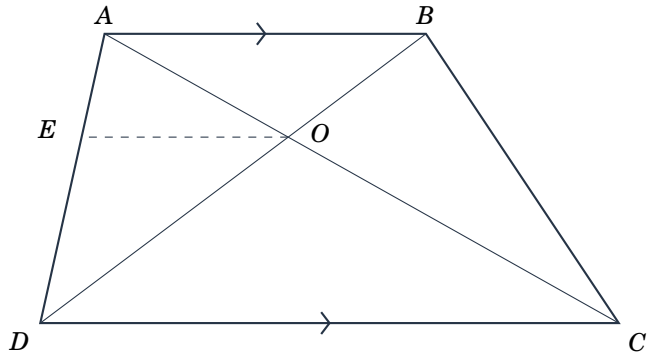
5. In $\triangle ABC$, D is on AB and E is on AC, with $AD = 4$ cm, $AE = 6$ cm, $DB = 8$ cm and $EC = 12$ cm. Sohan finds $\frac{AD}{AE} = \frac{DB}{EC} = \frac{2}{3}$. He says, "This is not the ratio in the converse of BPT, so I cannot decide whether $DE \parallel BC$." Is he right? Explain.

6. In $\triangle LMN$, X is on LM and Y is on LN. $LX = 3$ cm, $XM = 5$ cm, $LY = 6$ cm and $YN = 10$ cm. Leela says $XY \parallel MN$. Is she right? Explain.

7. Draw a triangle ABC with $AB = 6$ cm and $AC = 9$ cm. Mark D on AB with $AD = 2$ cm. Calculate where E must be on AC so that $DE \parallel BC$. Mark E, join DE, and check with a set square.

8. A carpenter fixes a crossbar DE on a triangular wooden frame ABC, with D on AB and E on AC. $AD = 25$ cm, $DB = 50$ cm, $AE = 30$ cm and $EC = 60$ cm. Will the crossbar be parallel to the base BC? Explain.

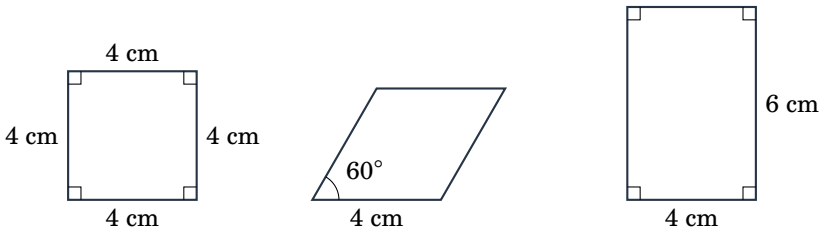
9. The diagonals of a quadrilateral $ABCD$ meet at O, and $\frac{AO}{BO} = \frac{CO}{DO}$. Prove that $ABCD$ is a trapezium. (Hint: draw $OE \parallel AB$, with E on AD.)



4 Why similarity criteria are needed

By definition, two triangles are similar when *all three* pairs of corresponding angles are equal *and* all three pairs of corresponding sides are in the same ratio. That is six checks. Do we always need all six?

For four-sided figures, one kind of check alone is not enough.



square rhombus rectangle

- The square and the rhombus have all sides in the same ratio, $\frac{4}{4} = 1$, but their angles differ. They do not have the same shape.
- The square and the rectangle have all angles equal, 90° each, but their sides are not in one ratio: $\frac{4}{4} \neq \frac{4}{6}$. They do not have the same shape.

Activity. Draw a triangle ABC with $BC = 4$ cm, $\angle B = 50°$ and $\angle C = 60°$. Draw a second triangle DEF with $EF = 6$ cm, $\angle E = 50°$ and $\angle F = 60°$.

(a) Find $\angle A$ and $\angle D$ without measuring.

(b) Measure AB, AC, DE and DF. Find $\frac{AB}{DE}$, $\frac{BC}{EF}$ and $\frac{AC}{DF}$.

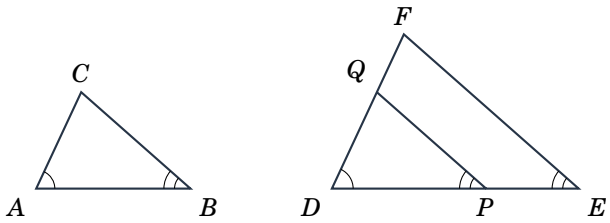
(c) What do you notice?

For triangles, the answer is different: a few of the six checks are enough. The rules that tell us which checks are enough are called **similarity criteria**. Sections 5, 6 and 7 give three of them.

5 Equal angles: the AA criterion

Theorem 3 (AAA similarity criterion)**.** *If in two triangles the corresponding angles are equal, then their corresponding sides are in the same ratio, and hence the two triangles are similar.*

Why it works. Let $\angle A = \angle D$, $\angle B = \angle E$ and $\angle C = \angle F$, with $\triangle DEF$ the larger triangle. Mark P on DE and Q on DF so that $DP = AB$ and $DQ = AC$.



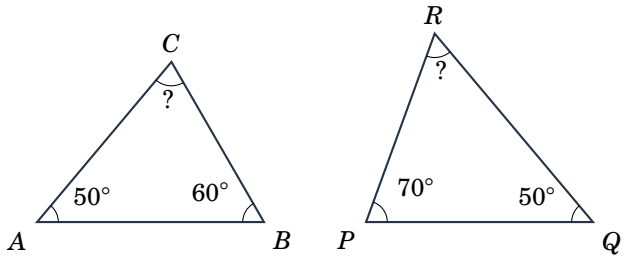
Then $\triangle ABC \cong \triangle DPQ$ (SAS congruence), so $\angle DPQ = \angle B = \angle E$. These are corresponding angles, so $PQ \parallel EF$. By BPT (part to whole side form), $\frac{DP}{DE} = \frac{DQ}{DF}$, that is, $\frac{AB}{DE} = \frac{AC}{DF}$. In the same way, $\frac{AB}{DE} = \frac{BC}{EF}$.

The angles of a triangle add up to 180°. So if two pairs of angles are equal, the third pair is equal too. This gives a shorter test.

Corollary 1 (AA similarity criterion)**.** *If two angles of one triangle are equal to two angles of another triangle, then the two triangles are similar.*

Writing the statement. Match the vertices with equal angles, and write them in the same order. If $\angle A = \angle Q$, $\angle B = \angle R$ and $\angle C = \angle P$, write $\triangle ABC \sim \triangle QRP$.

Example 5.1. In $\triangle ABC$, $\angle A = 50°$ and $\angle B = 60°$. In $\triangle PQR$, $\angle P = 70°$ and $\angle Q = 50°$. Are the triangles similar? If they are, write the similarity statement in the correct order.



Given: $\angle A = 50°$, $\angle B = 60°$, $\angle P = 70°$, $\angle Q = 50°$.
To find: whether the triangles are similar, and the similarity statement.

Solution.

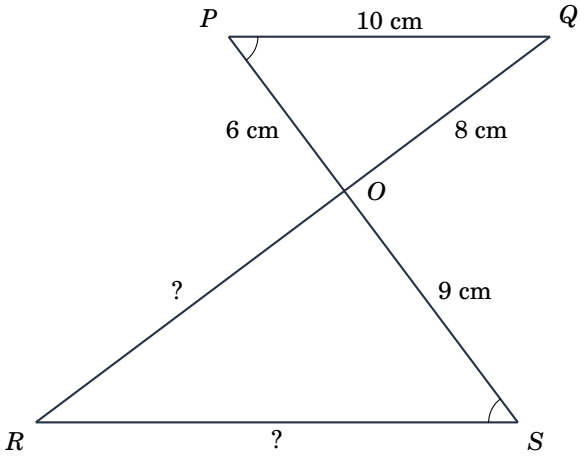
$$\angle C = 180° - 50° - 60° = 70° \quad \text{(angle sum of a triangle)}$$

$$\angle R = 180° - 70° - 50° = 60° \quad \text{(angle sum of a triangle)}$$

So $\angle A = \angle Q = 50°$, $\angle B = \angle R = 60°$ and $\angle C = \angle P = 70°$. Hence $A \leftrightarrow Q$, $B \leftrightarrow R$, $C \leftrightarrow P$.

Answer: Yes, by AA: $\triangle ABC \sim \triangle QRP$.

Example 5.2. $PQ \parallel RS$. The segments PS and QR meet at O. $OP = 6$ cm, $OS = 9$ cm, $OQ = 8$ cm and $PQ = 10$ cm. Prove that $\triangle POQ \sim \triangle SOR$. Then find OR and RS.



Given: $PQ \parallel RS$, $OP = 6$ cm, $OS = 9$ cm, $OQ = 8$ cm, $PQ = 10$ cm.
To prove: $\triangle POQ \sim \triangle SOR$. **To find:** OR and RS.

Solution.

$$\angle OPQ = \angle OSR \quad \text{(alternate angles, } PQ \parallel RS\text{)}$$

$$\angle POQ = \angle SOR \quad \text{(vertically opposite angles)}$$

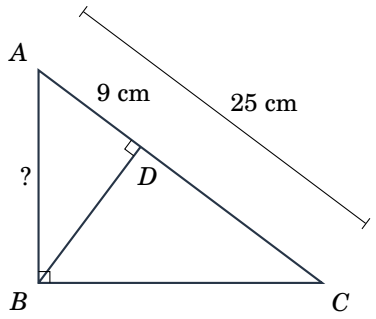
So $\triangle POQ \sim \triangle SOR$ (AA). Therefore

$$\frac{OP}{OS} = \frac{OQ}{OR} = \frac{PQ}{SR}, \qquad \frac{6}{9} = \frac{8}{OR} = \frac{10}{SR}.$$

So $OR = \dfrac{8 \times 9}{6} = 12$ and $SR = \dfrac{10 \times 9}{6} = 15$.

Answer: $OR = 12$ cm and $RS = 15$ cm.

Example 5.3. In $\triangle ABC$, $\angle ABC = 90°$. BD is perpendicular to AC, with D on AC. $AD = 9$ cm and $AC = 25$ cm. Prove that $\triangle ADB \sim \triangle ABC$. Then find AB.



Given: $\angle ABC = 90°$, $BD \perp AC$, $AD = 9$ cm, $AC = 25$ cm.
To prove: $\triangle ADB \sim \triangle ABC$. **To find:** AB.

Solution.

$$\angle ADB = \angle ABC = 90° \quad \text{(given)}$$

$$\angle BAD = \angle CAB \quad \text{(common angle)}$$

So $\triangle ADB \sim \triangle ABC$ (AA), with $A \leftrightarrow A$, $D \leftrightarrow B$, $B \leftrightarrow C$. Therefore

$$\frac{AD}{AB} = \frac{AB}{AC}, \qquad AB^2 = AD \times AC = 9 \times 25 = 225, \qquad AB = 15.$$

Answer: $AB = 15$ cm.

Exercise 2D

1. In $\triangle XYZ$, $\angle X = 40°$ and $\angle Y = 85°$. In $\triangle LMN$, $\angle L = 55°$ and $\angle N = 40°$. Are the triangles similar? If they are, write the similarity statement in the correct order.

2. Draw a triangle with a base of 4 cm and base angles 45° and 60°. Draw a second triangle with a base of 6 cm and the same base angles. Predict the ratio of the other two pairs of sides. Then measure and check.

3. $\triangle ABC \sim \triangle DEF$, with $AB = 8$ cm, $BC = 12$ cm and $DE = 6$ cm. Also $\angle A = 72°$ and $\angle E = 48°$. Find EF and $\angle F$.

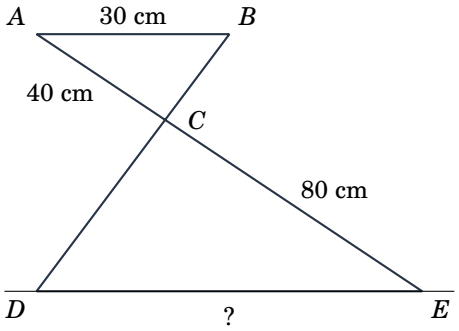
4. In $\triangle ABC$ and $\triangle PQR$, $\angle A = \angle P = 50°$ and $\angle B = \angle R = 70°$. A student writes $\triangle ABC \sim \triangle PQR$. Find the mistake and write the correct statement.

5. Meena makes two claims. Is each claim right? Explain.

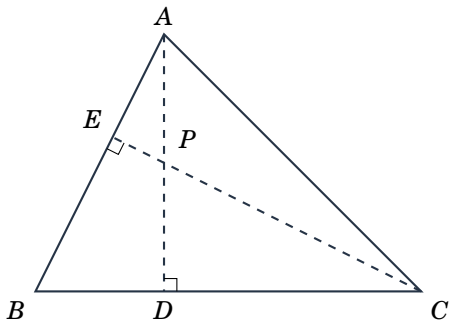
   (a) Any two equilateral triangles are similar.

   (b) Any two isosceles triangles are similar.

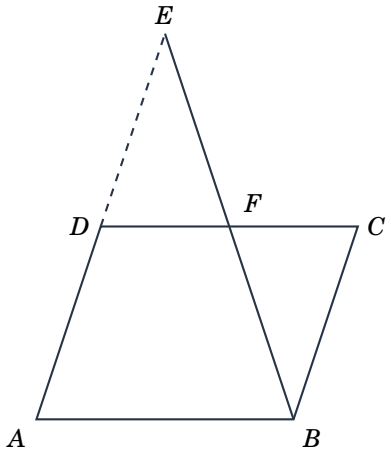
6. A folding stand for drying clothes has two straight legs AE and BD that cross at C. The top bar AB is parallel to the floor DE. $AC = 40$ cm, $CE = 80$ cm and $AB = 30$ cm. Find the distance DE between the feet of the stand.



7. In $\triangle ABC$, the altitudes AD and CE meet at P. Prove that $\triangle AEP \sim \triangle CDP$. Then prove that $\triangle ABD \sim \triangle CBE$.



8. $ABCD$ is a parallelogram. E is a point on AD produced beyond D. BE meets CD at F. Prove that $\triangle ABE \sim \triangle CFB$.



6 Proportional sides: the SSS criterion

Activity. (a) Use a compass and ruler to construct a triangle with sides 4 cm, 6 cm and 8 cm, and another with sides 6 cm, 9 cm and 12 cm. Find the ratio of the shortest sides, the middle sides and the longest sides.

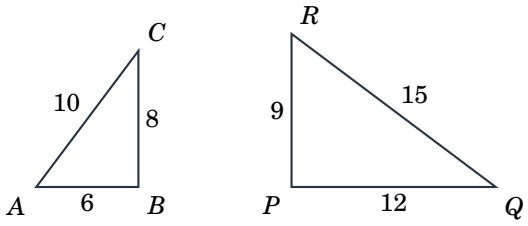
(b) Measure the angles of both triangles. What do you notice?

Theorem 4 (SSS similarity criterion)**.** *If in two triangles the sides of one triangle are proportional to the sides of the other triangle, then their corresponding angles are equal, and hence the two triangles are similar.*

Why it works. Let $\dfrac{AB}{DE} = \dfrac{BC}{EF} = \dfrac{CA}{FD}$, with $\triangle DEF$ the larger triangle. Mark P on DE and Q on DF so that $DP = AB$ and $DQ = AC$. Then $\dfrac{DP}{DE} = \dfrac{DQ}{DF}$, so $PQ \parallel EF$ (converse of BPT). So $\triangle DPQ \sim \triangle DEF$ (AA, using corresponding angles), and $\dfrac{PQ}{EF} = \dfrac{DP}{DE} = \dfrac{AB}{DE} = \dfrac{BC}{EF}$. Hence $PQ = BC$, and $\triangle DPQ \cong \triangle ABC$ (SSS congruence). So the angles of $\triangle ABC$ equal the angles of $\triangle DEF$.

Writing the statement. Match the shortest side with the shortest side, the middle with the middle, and the longest with the longest. Then the vertex where two matched sides meet corresponds to the vertex where their partners meet.

Example 6.1. In $\triangle ABC$, $AB = 6$ cm, $BC = 8$ cm and $CA = 10$ cm. In $\triangle PQR$, $PQ = 12$ cm, $QR = 15$ cm and $RP = 9$ cm. Are the triangles similar? If they are, write the similarity statement in the correct order.



Given: $AB = 6$, $BC = 8$, $CA = 10$; $PQ = 12$, $QR = 15$, $RP = 9$ (in cm).
To find: whether the triangles are similar, and the similarity statement.

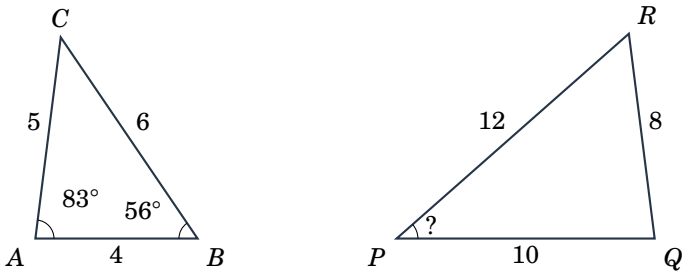
Solution. Arrange each triangle's sides from shortest to longest: AB, BC, CA and RP, PQ, QR.

$$\frac{AB}{RP} = \frac{6}{9} = \frac{2}{3}, \qquad \frac{BC}{PQ} = \frac{8}{12} = \frac{2}{3}, \qquad \frac{CA}{QR} = \frac{10}{15} = \frac{2}{3}.$$

All three ratios are equal, so the triangles are similar (SSS). AB and BC meet at B; their partners RP and PQ meet at P. So $B \leftrightarrow P$. In the same way $A \leftrightarrow R$ and $C \leftrightarrow Q$.

Answer: Yes, by SSS: $\triangle ABC \sim \triangle RPQ$.

Example 6.2. In $\triangle ABC$, $AB = 4$ cm, $BC = 6$ cm, $CA = 5$ cm, $\angle A = 83^\circ$ and $\angle B = 56^\circ$. In $\triangle PQR$, $PQ = 10$ cm, $QR = 8$ cm and $RP = 12$ cm. Find $\angle P$.



Given: $AB = 4$, $BC = 6$, $CA = 5$, $\angle A = 83^\circ$, $\angle B = 56^\circ$; $PQ = 10$, $QR = 8$, $RP = 12$ (in cm).
To find: $\angle P$.

Solution. Shortest to longest: AB, CA, BC and QR, PQ, RP.

$$\frac{AB}{QR} = \frac{4}{8} = \frac{1}{2}, \qquad \frac{CA}{PQ} = \frac{5}{10} = \frac{1}{2}, \qquad \frac{BC}{RP} = \frac{6}{12} = \frac{1}{2}.$$

So the triangles are similar (SSS). AB and CA meet at A; QR and PQ meet at Q. So $A \leftrightarrow Q$. In the same way $B \leftrightarrow R$ and $C \leftrightarrow P$. So $\triangle ABC \sim \triangle QRP$, and

$$\angle P = \angle C = 180^\circ - 83^\circ - 56^\circ = 41^\circ.$$

Answer: $\angle P = 41^\circ$.

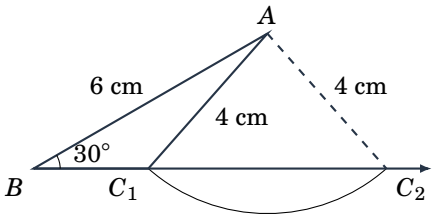
Exercise 2E

1. Decide whether each pair of triangles is similar. Give a reason.
   (a) Sides 3, 4, 6 cm and 6, 8, 12 cm.
   (b) Sides 5, 6, 8 cm and 10, 12, 15 cm.
   (c) Sides 2, 3, 4 cm and 9, 6, 12 cm.
2. The sides of $\triangle ABC$ are 6 cm, 9 cm and 12 cm. $\triangle DEF$ is similar to $\triangle ABC$, and its longest side is 20 cm. Find the other two sides of $\triangle DEF$.
3. In $\triangle ABC$, $AB = 4$ cm, $BC = 6$ cm and $CA = 8$ cm. In $\triangle PQR$, $PQ = 12$ cm, $QR = 9$ cm and $RP = 6$ cm. A student finds $\frac{AB}{PQ} = \frac{1}{3}$ and $\frac{BC}{QR} = \frac{2}{3}$, and says the triangles are not similar. Find the mistake. Are the triangles similar? If they are, write the similarity statement.
4. Rahul says, "A triangle with sides 5, 5, 8 cm and a triangle with sides 10, 10, 16 cm are similar." Is he right? Explain.
5. In $\triangle XYZ$, $XY = 3$ cm, $YZ = 5$ cm, $ZX = 7$ cm and $\angle Y = 120^\circ$. In $\triangle LMN$, $LM = 10$ cm, $MN = 6$ cm and $NL = 14$ cm. Write the similarity statement. Which angle of $\triangle LMN$ is 120°?
6. A school garden is a triangle with sides 45 m, 60 m and 75 m. Priya makes a model of it with sides 9 cm, 12 cm and 16 cm. Does her model have the same shape as the garden? If not, which side of the model should she change, and to what length?
7. D, E and F are the midpoints of the sides BC, CA and AB of $\triangle ABC$. Prove that $\triangle DEF \sim \triangle ABC$. (Use the midpoint theorem.)

7 Two sides and the included angle: the SAS criterion

Activity. (a) Draw $\triangle ABC$ with $AB = 3$ cm, $AC = 8$ cm and $\angle A = 60^\circ$. Draw $\triangle PQR$ with $PQ = 6$ cm, $PR = 16$ cm and $\angle P = 60^\circ$. Measure BC and QR. Find $\frac{AB}{PQ}$, $\frac{AC}{PR}$ and $\frac{BC}{QR}$.

(b) Now draw a triangle with $AB = 6$ cm, $\angle B = 30^\circ$ and $AC = 4$ cm. Can you draw two different triangles with these three measurements?

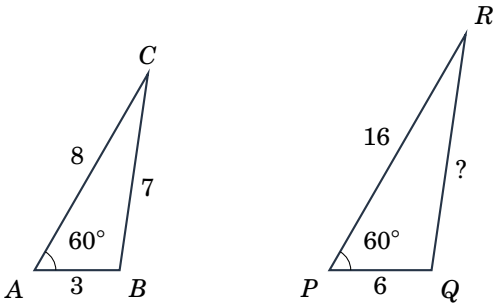


In part (b) the angle B is *not* between the two given sides AB and AC. Two different shapes are possible, so these measurements do not fix the shape. In part (a) the angle is between the two sides. This angle is called the **included angle**.

Theorem 5 (SAS similarity criterion)**.** *If one angle of a triangle is equal to one angle of the other triangle and the sides including these angles are proportional, then the two triangles are similar.*

Why it works. Let $\angle A = \angle D$ and $\dfrac{AB}{DE} = \dfrac{AC}{DF}$, with $\triangle DEF$ the larger triangle. Mark P on DE and Q on DF so that $DP = AB$ and $DQ = AC$. Then $\dfrac{DP}{DE} = \dfrac{DQ}{DF}$, so $PQ \parallel EF$ (converse of BPT), and $\triangle DPQ \sim \triangle DEF$ (AA, using corresponding angles). Also $\triangle DPQ \cong \triangle ABC$ (SAS congruence). Hence $\triangle ABC \sim \triangle DEF$.

Example 7.1. In $\triangle ABC$, $AB = 3$ cm, $AC = 8$ cm, $\angle A = 60^\circ$ and $BC = 7$ cm. In $\triangle PQR$, $PQ = 6$ cm, $PR = 16$ cm and $\angle P = 60^\circ$. Prove that $\triangle ABC \sim \triangle PQR$. Then find QR.



Given: $AB = 3$, $AC = 8$, $\angle A = 60^\circ$, $BC = 7$; $PQ = 6$, $PR = 16$, $\angle P = 60^\circ$ (in cm).
To prove: $\triangle ABC \sim \triangle PQR$. **To find:** QR.

Solution.

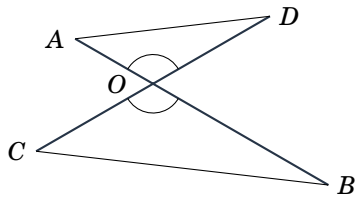
$$\frac{AB}{PQ} = \frac{3}{6} = \frac{1}{2}, \quad \frac{AC}{PR} = \frac{8}{16} = \frac{1}{2} \quad \text{(sides including } \angle A \text{ and } \angle P\text{)}$$

$$\angle A = \angle P = 60^\circ \quad \text{(given)}$$

So $\triangle ABC \sim \triangle PQR$ (SAS). Then $\dfrac{BC}{QR} = \dfrac{1}{2}$, so $QR = 2 \times 7 = 14$.

Answer: $QR = 14$ cm.

Example 7.2. The segments AB and CD meet at O, with O between A and B and between C and D. $OA \times OB = OC \times OD$. Prove that $\angle OAD = \angle OCB$ and $\angle ODA = \angle OBC$.



Given: $OA \times OB = OC \times OD$.
To prove: $\angle OAD = \angle OCB$ and $\angle ODA = \angle OBC$.

Proof.

$$\frac{OA}{OC} = \frac{OD}{OB} \quad \text{(divide both sides of the given equation by } OC \times OB\text{)}$$

$$\angle AOD = \angle COB \quad \text{(vertically opposite angles)}$$

In $\triangle AOD$ and $\triangle COB$, the sides including these equal angles are proportional. So $\triangle AOD \sim \triangle COB$ (SAS), with $A \leftrightarrow C$, $O \leftrightarrow O$, $D \leftrightarrow B$. Therefore $\angle OAD = \angle OCB$ and $\angle ODA = \angle OBC$. □

Exercise 2F

1. In $\triangle PQR$, $PQ = 5$ cm, $PR = 8$ cm and $\angle P = 40°$. In $\triangle XYZ$, $XY = 10$ cm, $XZ = 16$ cm and $\angle X = 40°$. Are the triangles similar? If they are, write the similarity statement and find the ratio $QR : YZ$.

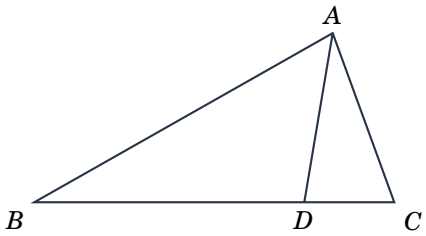
2. In $\triangle ABC$, $AB = 6$ cm, $BC = 9$ cm and $\angle A = 50°$. In $\triangle DEF$, $DE = 4$ cm, $EF = 6$ cm and $\angle D = 50°$. Can you conclude that the triangles are similar? Explain.

3. In $\triangle ABC$, $AB = 8$ cm, $AC = 12$ cm and $\angle A = 70°$. In $\triangle DEF$, $DE = 6$ cm and $\angle D = 70°$. Find the length DF must have for $\triangle ABC \sim \triangle DEF$.

4. A student writes: "$\dfrac{AB}{DE} = \dfrac{BC}{EF} = 2$ and $\angle A = \angle D$. So $\triangle ABC \sim \triangle DEF$ by SAS." Find the mistake. What should be given instead of $\angle A = \angle D$?

5. In $\triangle ABC$, D is on AB and E is on AC. $AD = 3$ cm, $AB = 9$ cm, $AE = 4$ cm and $AC = 12$ cm. Anu says, "$\triangle ADE \sim \triangle ABC$, and so $DE \parallel BC$." Is she right? Explain.

6. D is a point on the side BC of $\triangle ABC$ such that $CA^2 = CB \times CD$. Prove that $\triangle ADC \sim \triangle BAC$. Hence prove that $\angle CAD = \angle CBA$.



7. The frame for a festival stage has two straight poles, 3 m and 8 m long, fixed at one end with an angle of 60° between them. A rope of 7 m ties their free ends. A model of the frame uses sticks of 15 cm and 40 cm at the same angle. Find the length of the tie in the model.

8 Choosing the criterion

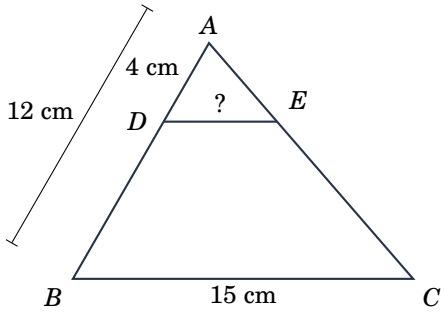
Criterion	What must be known	Watch for
AA	two pairs of equal angles	parallel lines, vertically opposite angles, a common angle, right angles
SSS	three pairs of sides in one ratio	sort the sides by length before pairing
SAS	two pairs of sides in one ratio, and the angles *between* them equal	the angle must be included

A proof of similarity is written in four steps.

1. Name the two triangles.
2. List only the facts the criterion needs, with a reason for each.
3. Name the criterion and write the similarity statement in matching order.

4. Only then use the equal angles or the proportional sides.

Example 8.1. In $\triangle ABC$, $DE \parallel BC$, with D on AB and E on AC. $AD = 4$ cm, $AB = 12$ cm and $BC = 15$ cm. Find DE.



Given: $DE \parallel BC$, $AD = 4$ cm, $AB = 12$ cm, $BC = 15$ cm.
To find: DE.

Solution. BPT compares only the pieces of AB and AC. For DE we need similar triangles.

$$\angle ADE = \angle ABC \quad \text{(corresponding angles, } DE \parallel BC\text{)}$$
$$\angle DAE = \angle BAC \quad \text{(common angle)}$$

So $\triangle ADE \sim \triangle ABC$ (AA). Therefore

$$\frac{DE}{BC} = \frac{AD}{AB}, \qquad \frac{DE}{15} = \frac{4}{12}, \qquad DE = 5.$$

Answer: $DE = 5$ cm.

Exercise 2G

For each case, name the criterion that proves the two triangles similar, or write "not enough information". Give a reason.

1. $\angle A = \angle P$ and $\angle B = \angle Q$.

2. The sides are 4, 6, 8 cm and 6, 9, 12 cm.

3. $\dfrac{AB}{PQ} = \dfrac{AC}{PR} = \dfrac{2}{3}$ and $\angle B = \angle Q$.

4. $\dfrac{AB}{PQ} = \dfrac{AC}{PR} = \dfrac{5}{7}$ and $\angle A = \angle P$.

5. Only $\angle A = \angle P$ is known.

6. Both triangles are right-angled, and one acute angle of the first equals one acute angle of the second.

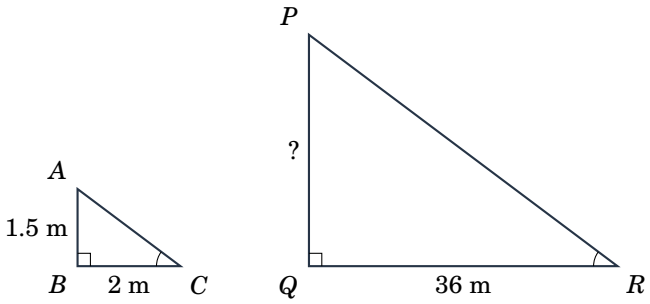
7. In $\triangle ABC$, $DE \parallel BC$, with D on AB and E on AC; the triangles are $\triangle ADE$ and $\triangle ABC$.

8. Both triangles are right-angled. The sides forming the right angle are 5 cm and 12 cm in one, and 10 cm and 24 cm in the other.

9. Two triangles have angles 48°, 62°, 70° and 70°, 48°, 62°, but very different side lengths.

9 Using similarity

Example 9.1. At the same time of day, a boy 1.5 m tall casts a shadow 2 m long, and a temple tower casts a shadow 36 m long. Find the height of the tower.



Given: boy $AB = 1.5$ m, his shadow $BC = 2$ m; tower shadow $QR = 36$ m.
To find: the height PQ of the tower.

Solution.

$$\angle ABC = \angle PQR = 90^\circ \quad \text{(both stand vertically on level ground)}$$

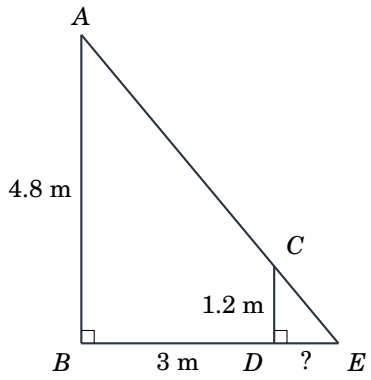
$$\angle ACB = \angle PRQ \quad \text{(the sun's rays make the same angle with the ground)}$$

So $\triangle ABC \sim \triangle PQR$ (AA). Therefore

$$\frac{PQ}{AB} = \frac{QR}{BC}, \qquad \frac{PQ}{1.5} = \frac{36}{2}, \qquad PQ = 1.5 \times 18 = 27.$$

Answer: The tower is 27 m high.

Example 9.2. A girl 1.2 m tall walks away from the base of a lamp post at 1 m per second. The lamp is 4.8 m above the ground. Find the length of her shadow after 3 seconds.



Given: $AB = 4.8$ m, $CD = 1.2$ m, $BD = 1 \times 3 = 3$ m.
To find: the shadow DE.

Solution. Let $DE = s$ m.

$$\angle EDC = \angle EBA = 90^\circ \quad \text{(both vertical on level ground)}$$

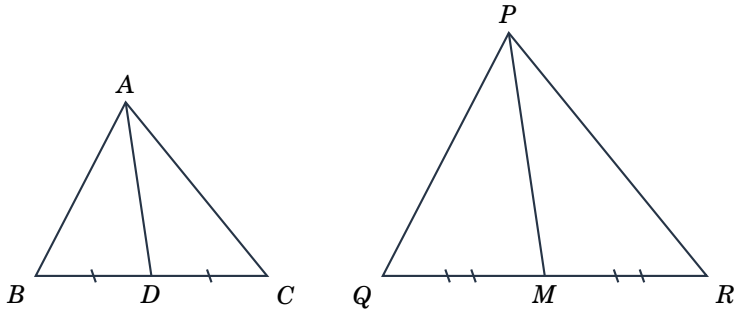
$$\angle DEC = \angle BEA \quad \text{(common angle at } E\text{)}$$

So $\triangle EDC \sim \triangle EBA$ (AA). Therefore

$$\frac{DE}{BE} = \frac{CD}{AB}, \qquad \frac{s}{s+3} = \frac{1.2}{4.8} = \frac{1}{4}, \qquad 4s = s + 3, \qquad s = 1.$$

Answer: Her shadow is 1 m long.

Example 9.3. $\triangle ABC \sim \triangle PQR$. AD and PM are medians of the two triangles. Prove that $\dfrac{AB}{PQ} = \dfrac{AD}{PM}$.



Given: $\triangle ABC \sim \triangle PQR$; D is the midpoint of BC and M is the midpoint of QR.
To prove: $\dfrac{AB}{PQ} = \dfrac{AD}{PM}$.

Proof.

$$\begin{aligned} \frac{AB}{PQ} &= \frac{BC}{QR} && \text{(corresponding sides, } \triangle ABC \sim \triangle PQR) \\ &= \frac{2BD}{2QM} = \frac{BD}{QM} && (D \text{ and } M \text{ are midpoints}) \\ \angle ABD &= \angle PQM && \text{(corresponding angles, } \triangle ABC \sim \triangle PQR) \end{aligned}$$

So $\triangle ABD \sim \triangle PQM$ (SAS). Therefore $\dfrac{AB}{PQ} = \dfrac{AD}{PM}$. □

Exercise 2H

1. In $\triangle ABC$, $DE \parallel BC$, with D on AB and E on AC. $AD = 4$ cm, $DB = 8$ cm and $BC = 18$ cm. Divya writes $\dfrac{DE}{BC} = \dfrac{AD}{DB} = \dfrac{1}{2}$, so $DE = 9$ cm. Find her mistake and find the correct DE.

2. $ABCD$ is a trapezium with $AB \parallel DC$. Its diagonals meet at O. $AO = 6$ cm, $OC = 9$ cm, $BO = 8$ cm and $AB = 10$ cm. Draw a rough figure and mark the given lengths. Find OD and DC.

3. A straight stick 2 m long stands upright on the level school ground and casts a shadow 1.5 m long. At the same time the flagpole casts a shadow 6 m long.

   (a) Find the height of the flagpole.

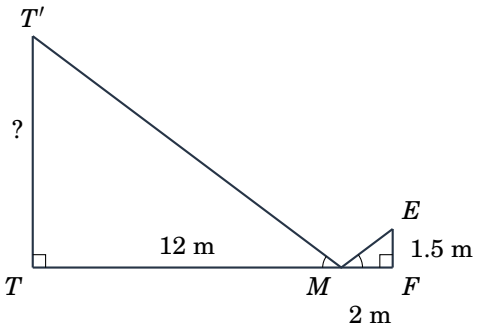
   (b) Later in the day the flagpole's shadow is 10 m long. Find the length of the stick's shadow at that time.

4. A man 1.8 m tall stands 4 m from the base of a street lamp 5.4 m high. Draw a rough figure and mark the given lengths.

   (a) Find the length of his shadow.

   (b) What if he moves 2 m farther from the lamp? Find his new shadow.

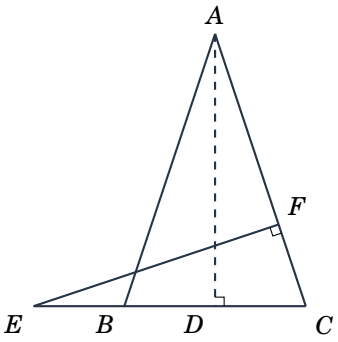
5. To find the height of a coconut tree, Arun lays a small mirror flat on level ground 12 m from the foot of the tree. He stands 2 m from the mirror, on the other side, and sees the top of the tree in the mirror. His eyes are 1.5 m above the ground. The angle at which light falls on the mirror equals the angle at which it leaves the mirror. Find the height of the tree.



6. $\triangle ABC \sim \triangle PQR$, with $AB = 12$ cm and $PQ = 8$ cm. The median AD of $\triangle ABC$ is 9 cm. Find the median PM of $\triangle PQR$.

7. $\triangle ABC \sim \triangle PQR$. AD and PM are the altitudes of the two triangles, with D on BC and M on QR. Prove that $\dfrac{AB}{PQ} = \dfrac{AD}{PM}$.

8. In the isosceles triangle ABC, $AB = AC$. E is a point on CB produced beyond B. $AD \perp BC$, with D on BC, and $EF \perp AC$, with F on AC. Prove that $\triangle ABD \sim \triangle ECF$.



9. In Question 2, what if AB were equal to DC? What kind of quadrilateral would $ABCD$ be, and how would AO compare with OC?

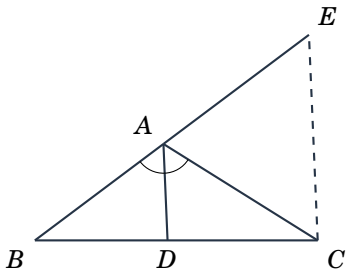
Things to Remember

- **BPT:** if $DE \parallel BC$ in $\triangle ABC$, then $\frac{AD}{DB} = \frac{AE}{EC}$, and also $\frac{AD}{AB} = \frac{AE}{AC}$.
- **Converse of BPT:** if $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$.
- **AA:** two pairs of equal angles. **SSS:** three pairs of sides in one ratio. **SAS:** two pairs of sides in one ratio with the *included* angles equal.
- Two sides in one ratio and an angle that is *not* included do not prove similarity.
- Write the similarity statement so that corresponding vertices are in the same order.
- For a length such as DE, use similar triangles: $\frac{DE}{BC} = \frac{AD}{AB}$, not $\frac{AD}{DB}$.

10 If you are ready for more

Exercise R

1. Use BPT to prove: a line drawn through the midpoint of one side of a triangle, parallel to another side, bisects the third side.
2. In $\triangle ABC$, AD bisects $\angle A$ and meets BC at D. Prove that $\frac{BD}{DC} = \frac{AB}{AC}$. (Hint: through C, draw a line parallel to DA to meet BA produced at E.)



3. Can two triangles have all three angles of one equal to the three angles of the other, and two sides of one equal to two sides of the other, and still not be congruent? Test the triangles with sides 8, 12, 18 cm and 12, 18, 27 cm.
4. In $\triangle ABC$ and $\triangle PQR$, AD and PM are medians, and $\frac{AB}{PQ} = \frac{AC}{PR} = \frac{AD}{PM}$. Prove that $\triangle ABC \sim \triangle PQR$. (Hint: produce AD to E so that $DE = AD$, and PM to N so that $MN = PM$.)